

IV. Exogenous nutrition

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Protein turnover and amino acid flux in fish larvae

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The aim of this paper is to present a quantitative analysis of the amino acid flux in fish larvae. The important parts of the analysis are: food intake rates, assimilation rates, rates of protein synthesis, growth and degradation, and ammonia excretion rates. The greatest degree of uncertainty concerns the rates of protein synthesis in fish larvae. However, quantification of amino acid incorporation rates into protein is important when considering amino acid requirements for growth. Methods are described for measuring protein synthesis rates in small larval and juvenile fish based upon the phenylalanine flooding methodology which has been used successfully for larger fish. In trout fry, bathing the animals in radiolabelled amino acids is not very successful in terms of equilibrating the free amino acid pool with the bathing medium, but this method does work well for marine fish larvae and juvenile freshwater stages that are feeding. Results from injecting small trout fry with radiolabelled amino acids are described and this method works well for larger juvenile fish. From the protein synthesis rates of larval herring, an amino acid flux model is described which emphasizes the relatively large size of the free amino acid pool compared with the amino acids in proteins, and the size of the free pool is compared with estimated rates of protein synthesis. Other methods are described for determining the rates of protein turnover in fish larvae. The effects of body size are considered and it is concluded that RNA concentration, free amino acid pool, and the relative rates of protein synthesis and growth are not exceptional for fish larvae and seem to obey scaling relationships familiar from metabolic considerations. If protein synthesis accounts for a large proportion of the oxygen consumption in fish larvae, then the proportion of the synthesized protein retained as growth may play an important role in the growth efficiency (growth rate divided by food consumption), but there is no evidence that small fish larvae sacrifice protein turnover during rapid growth in order to reduce energy expenditure.

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Introduction

Protein turnover can be divided into its constituent processes, protein synthesis, protein accretion, and protein degradation (reviewed by Houlihan, 1991). These rates are commonly expressed on a fractional basis – the percentage of the body protein mass synthesized (k_s), accreted (k_g), or degraded (k_d) per day. At any particular time, protein growth is the net balance between protein synthesis and protein degradation, i.e., $k_g = k_s - k_d$ (Millward *et al.*, 1975, 1976). Rates of protein growth

are mainly determined by rates of protein ingestion, protein growth is some fraction of the rate of protein synthesis and there has been some speculation about the relationship between rates of protein ingestion and protein synthesis (Millward, 1989; Houlihan *et al.*, 1993a).

From these simple statements some questions can be posed about the physiological influences determining rates of protein turnover in small, rapidly growing animals such as larval or juvenile fish. In these animals, high weight-specific growth rates may be achieved by relatively low rates of protein turnover (equivalent to pro-

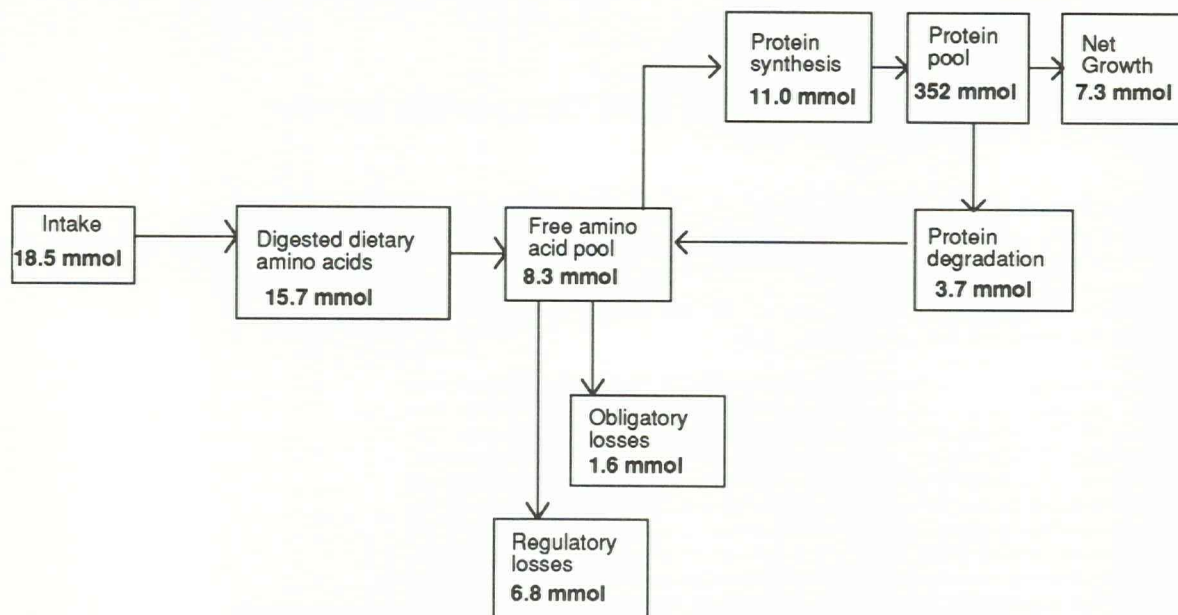


Figure 1. Amino acid flux for 250 g rainbow trout at 10°C eating a daily ration of 2% body weight d^{-1} based on the model of Millward and Rivers (1988). The protein content of the diet is 41% and the apparent absorption efficiency is assumed to be 85% (Kaushik and Gomes, 1988). One gramme of protein is equal to 9 mmol of free amino acids.

tein degradation in growing animals, Wiesner and Zak, 1991) i.e., rates of protein synthesis and protein growth may be roughly similar and degradation rates comparatively low. Alternatively, protein synthesis rates may be very much higher than protein growth rates, resulting in high rates of protein turnover (e.g., Fauconneau *et al.*, 1986a, b).

These alternative scenarios may have important implications for larval survival, as it has been postulated that high levels of protein turnover may increase viability and survival through the processes of repair (e.g., Kirkwood, 1981; Calow, 1985). However, the critical importance of protein turnover has also been increasingly linked to whole-animal energetics through the high cost of protein synthesis (e.g., Jobling, 1985; Houlihan, 1991). In small fish it has been suggested that the switching of energy allocation between maintenance and growth is particularly critical (e.g., Wieser *et al.*, 1992). A final question can be posed about amino acids and protein synthesis, as it is believed that for optimal (or maximal) rates of protein synthesis all essential amino acids must be presented simultaneously at the appropriate concentrations to the tissues (Tacon and Cowey, 1985). However, we know very little about acceptable variations in amino acid concentration in tissues and how high or low proportions of individual amino acids may influence rates of protein synthesis in larval fish as distinct from rates of protein growth.

Thus, there are a number of reasons for investigating protein turnover and amino acid flux in fish larvae. One approach, which also takes into consideration the nutritional aspect, is to use the amino acid and protein metabolism model of Millward and Rivers (1988) and to adapt it for fish larvae and juveniles. We have used recent data from large rainbow trout (*Oncorhynchus mykiss* Walbaum) in order to give some insight into how larval fish may differ from larger fish (Fig. 1). We have set the parameters for a 250 g live weight rainbow trout eating a daily ration of 2% of its body weight per day; the diet contained 42% protein and the digestive efficiency can be taken to be 85% (Kaushik and Gomes, 1988). Thus, 15.7 mmoles of amino acids would enter the free amino acid pool on a daily basis. Estimates from individual tissues indicate that the free amino acid pool, at 8.3 mmoles, represents only 2 to 3% of the total amount of amino acids in the whole animal (i.e., free amino acids plus the amino acids in body protein). Therefore, the amino acids absorbed from the daily food consumption represent a potential twofold expansion of the whole-animal free amino acid pool. However, for fish that have been fed regularly the evidence is that the tissue-free pools remain relatively stable in concentration (McCarthy, 1993; Carter *et al.*, 1995) and therefore the rates of protein synthesis and amino acid oxidation can be seen as homeostatic mechanisms maintaining tissue-free amino acid pool concentrations.

Table 1. Time course of free pool (s_a) and protein-bound (s_b) phenylalanine-specific radioactivity and fractional rates of protein synthesis (k_s) for rainbow trout yolk-sac fry. The radiolabel was administered to the fish by (a) bathing the fish in a 2% solution of 135 mM phenylalanine/L-[2, 6- 3 H]-phenylalanine (McCarthy *et al.*, unpubl.) or (b) following a single injection of 135 mM phenylalanine/L-[2, 6- 3 H]-phenylalanine into the yolk sac (McCarthy *et al.*, unpubl.). The data are presented as mean $\pm$ s.e. in parentheses for each time interval.

(a) Bathing method					
Incubation time (h)	2	4	6	9	12
No. of fish	10	10	10	10	9
Wet weight (mg) ^a	127 (4)	139 (8)	130 (17)	131 (5)	131 (5)
s_a (dpm nmol ⁻¹)	109 (11)	137 (5)	184 (14)	261 (22)	220 (37)
s_b (dpm nmol ⁻¹)	3.7 (0.1)	5.4 (0.7)	6.4 (0.4)	7.0 (0.6)	6.0 (1.8)
k_s (% d ⁻¹)	390 (4)	178 (3)	115 (1)	76 (0.5)	61 (1)
(b) Injection method					
Incubation time (h)	0.5	1	2		
No. of fish	9	12	10		
Wet weight (mg) ^b	104 (4)	109 (8)	117 (17)		
s_a (dpm nmol ⁻¹)	199 (59)	395 (22)	281 (42)		
s_b (dpm nmol ⁻¹)	1.0 (0.2)	2.1 (0.2)	2.8 (0.3)		
k_s (% d ⁻¹)	25 (6)	13 (0.7)	15 (2.9)		

a = body plus yolk sac.

b = body only.

In order to construct a similar model for larval fish some quantitative values are needed, particularly on rates of protein synthesis. Estimates and data are available on fish larval food consumption rates and digestive efficiency (Kjørboe, 1989) and recently much more has become known about amino acid levels in marine fish eggs and larvae (Rønnestad, 1992). Therefore, the primary aim of this paper is to review the methods for measuring protein synthesis rates in larval and small juvenile fish, and to discuss the results obtained and their significance in relation to larger fish. Because of the uncertainties in protein synthesis measurements in small larvae the effects of body size on protein synthesis and growth have been examined in order to find out whether small animals fall on a continuum with larger fish. This hypothesis was also tested for whole-body nucleic acid concentrations and ratios and whole-body free amino acid pool concentrations.

Rates of protein synthesis in larval fish have been measured *in vitro* by the incorporation of radiolabelled 14 C-phenylalanine into isolated polyribosomes in cod (*Gadus morhua* L.) (Hansen *et al.*, 1989). This approach may prove to be very valuable in estimating maximum rates of protein synthesis of larval and small juvenile fish. However, it is difficult to compare incorporation rates of radiolabelled amino acid of isolated ribosomes with rates of protein growth (k_g) and hence to calculate protein degradation rates (k_d). Protein synthesis rates *in vivo* of larval or juvenile fish have been measured by bathing the fish in water containing 14 C-arginine (Fauconneau, 1984; Fauconneau *et al.*, 1986a, b), or 3 H-phenylalanine (Houlihan *et al.*, 1992, 1993b, 1995), or by microinjection of 3 H-phenylalanine (McCarthy, 1993; Mathers *et al.*, 1993). These *in vivo* techniques

have allowed rates of protein synthesis to be measured in fish between 10 and 500 mg wet weight.

Measurements of protein synthesis rates in larval fish have given contradictory results. Early measurements using the bathing method of Fauconneau (1984) suggested that rates of protein synthesis are very high relative to growth rate, indicating very high rates of protein turnover. Early studies on carp, *Cyprinus carpio* L. (10 mg wet weight, 26°C, Fauconneau, 1984), whitefish, *Coregonus schinzi palea* Cuv. et Val. (60 mg wet weight, 13°C, Fauconneau *et al.*, 1986a) and sturgeon, *Acipenser transmontanus* Brandt (180 mg wet weight, 17°C, Fauconneau *et al.*, 1986b) indicated that the efficiency with which synthesized protein was retained as growth (growth rates divided by synthesis rates expressed as a percentage) was between 5 and 9%. However, more recently synthesis retention efficiencies of 40 to 50% have been reported for larval nase, *Chondrostoma nasus* L. (20–45 mg wet weight, 20°C, Houlihan *et al.*, 1992), and juvenile tilapia, *Oreochromis mossambicus* L. (20–100 mg wet weight, 26°C, Houlihan *et al.*, 1993b). These efficiencies are similar or only slightly higher than the values found in larger fish. In the following section the two methods that have so far been developed to measure *in vivo* rates of protein synthesis in small larval and juvenile fish will be described.

Methods for measuring protein synthesis rates of small larval and juvenile fish

Bathing in radiolabelled amino acids

Rainbow trout yolk-sac fry (131 mg wet weight, 10°C, McCarthy *et al.*, unpubl.) were placed in 980 ml of fresh

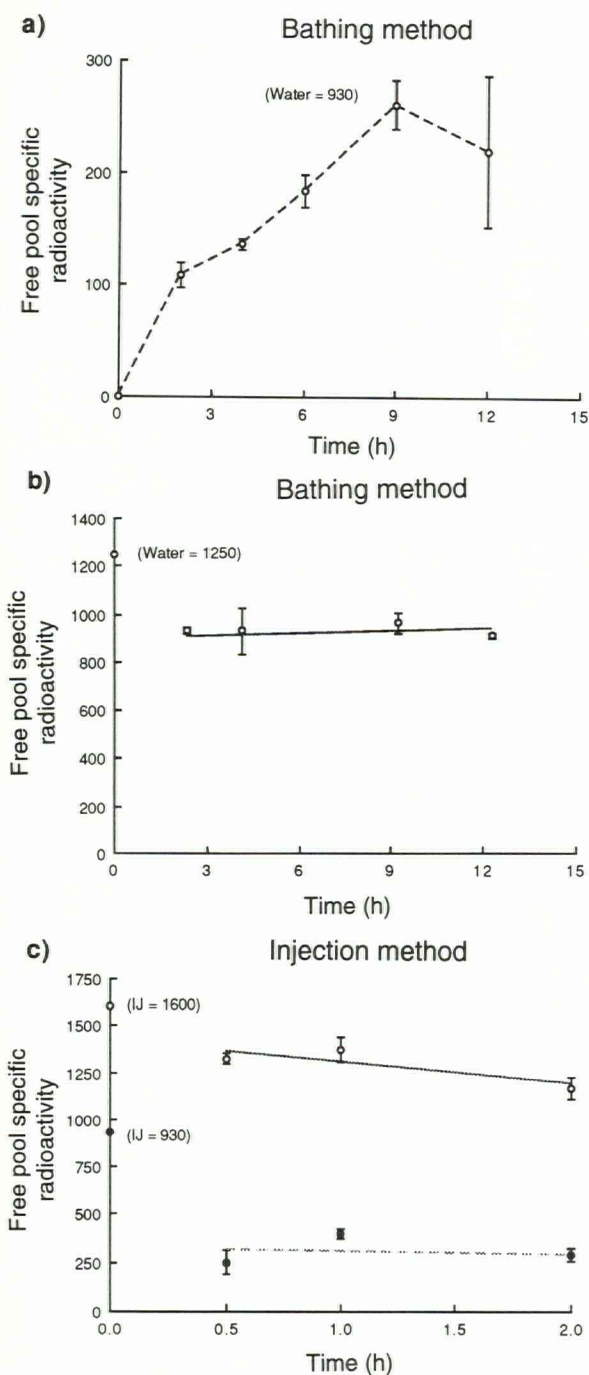


Figure 2. Time course of free phenylalanine-specific radioactivity (dpm nmol⁻¹) in (a) rainbow trout yolk-sac fry (131 mg, 8°C) placed in a 2% solution of 135 mM phenylalanine/L-[2,6-³H]-phenylalanine (McCarthy *et al.*, unpubl.), (b) larval herring (0.08 mg dry weight, 8°C) placed in a 2% solution of 135 mM phenylalanine/L-[2,6-³H]-phenylalanine (Houlihan *et al.*, 1995), and (c) rainbow trout yolk-sac fry (110 mg, 10°C, closed circles, McCarthy *et al.*, unpubl.) and rainbow trout fry (203 mg, 5°C, open circles, Mathers *et al.*, 1993) following a single injection of 135 mM phenylalanine/L-[2,6-³H]-phenyl-

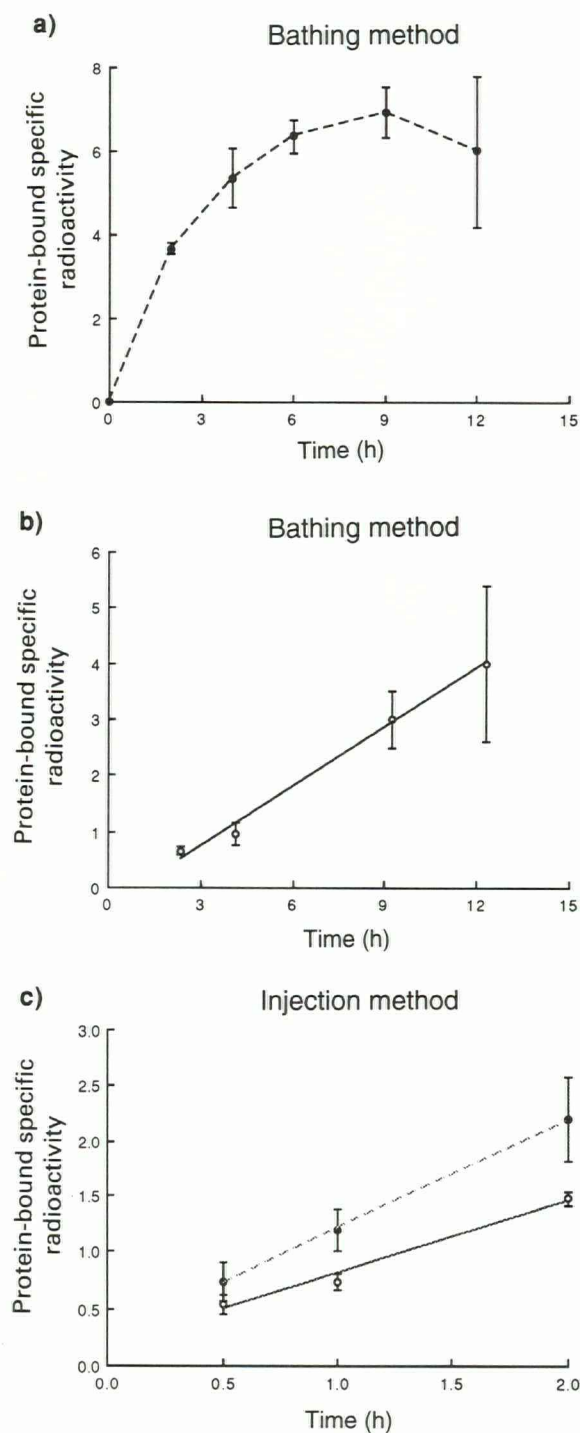


Figure 3. Time course of whole-animal protein-bound phenylalanine specific radioactivity (dpm nmol⁻¹) in the rainbow trout yolk-sac fry (a), larval herring (b), and rainbow trout yolk-sac fry and fry (c) described in Figure 2.

alanine into the yolk sac or the peritoneal cavity. The specific radioactivity of the bathing solution (a, b) and the injection solution (c) is indicated in parentheses.

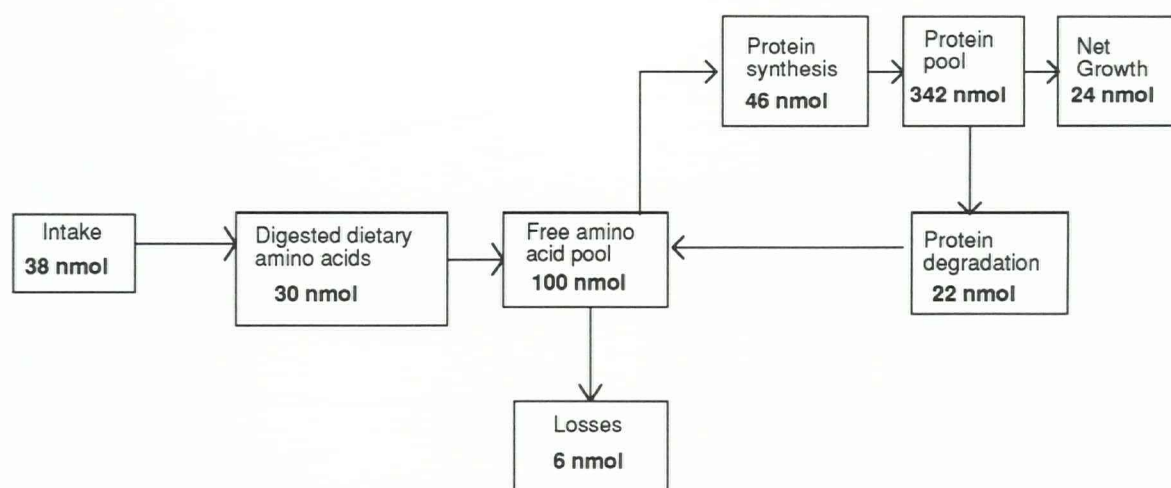


Figure 4. Amino acid flux for herring larva (0.08 mg dry weight, 8°C) eating a daily ration of 11% body weight d^{-1} based on the model of Millward and Rivers (1988). The apparent absorption efficiency is assumed to be 79% (Kjørboe, 1989). One gramme of protein is equal to 9 mmol of free amino acids.

Table 2. Whole-animal RNA efficiencies (k_{RNA} , g protein synthesized per day per unit of cellular RNA) for fed and starved fish. The wet weight (g), water temperature (°C), and the number of fish involved in the calculation are indicated. The data are presented as mean $\pm$ s.e.

Species		Weight	°C	k_{RNA}	($\pm$ s.e.)	n	Reference
Rainbow trout fry	Fed	0.203	5	0.85	0.10	15	Mathers <i>et al.</i> (1993)
	Starve	0.144	5	0.48	0.03	10	
	Fed	0.292	10	2.45	0.10	10	
	Starve	0.127	10	0.63	0.04	9	
	Fed	0.314	15	2.30	0.17	12	
	Starve	0.112	15	0.88	0.11	5	
Grass carp	Fed	23	22	5.93	0.30	15	Carter <i>et al.</i> (1993)
	Starve	15	22	1.87	0.40	6	
Atlantic salmon		175	14	2.25	0.31	10	Carter <i>et al.</i> (unpubl.)
		200	14	3.78	0.86	5	
Rainbow trout	Fed	80	8	3.94	0.30	27	McCarthy <i>et al.</i> (unpubl.)
	Starve	40	8	2.41	0.19	10	
Rainbow trout	Fed	300	14	3.13	0.33	15	
	Starve	170	14	1.96	0.22	8	

water. Twenty ml of 135 mM phenylalanine (pH 8) containing L-[2, 6- 3H] phenylalanine (3.6 MBq ml^{-1} , Amersham International) was prepared and added to the water. The final molarity of the solution was 2.7 mM and the phenylalanine specific radioactivity was measured as 930 disintegrations $minute^{-1}$ nmole phenylalanine $^{-1}$ (dpm nmole $^{-1}$). Air was gently bubbled into the beaker to oxygenate and mix the incubation solution and fish were removed after 2, 4, 6, 9, and 12 h of incubation. The fish were rinsed thoroughly with distilled water, blotted dry and killed by destruction of the brain and transection of the spinal cord. They were weighed, frozen in liquid nitrogen and stored at $-70^{\circ}C$

until further analysis. The free pool and protein-bound phenylalanine-specific radioactivity for each fish was determined as described by Houlihan *et al.* (1986, 1988) and the mean values for each time interval are given in Table 1. The free-pool phenylalanine-specific radioactivity reached a plateau of 240 dpm nmole $^{-1}$ after approximately 9 h incubation (Fig. 2a). However, the phenylalanine-specific radioactivity of the water was considerably higher (930 dpm nmole $^{-1}$) and the phenylalanine-specific radioactivity of the free pool was 26% of that of the bathing solution.

The extent of flooding of the amino acid free pool after bathing can be estimated from the concentration of

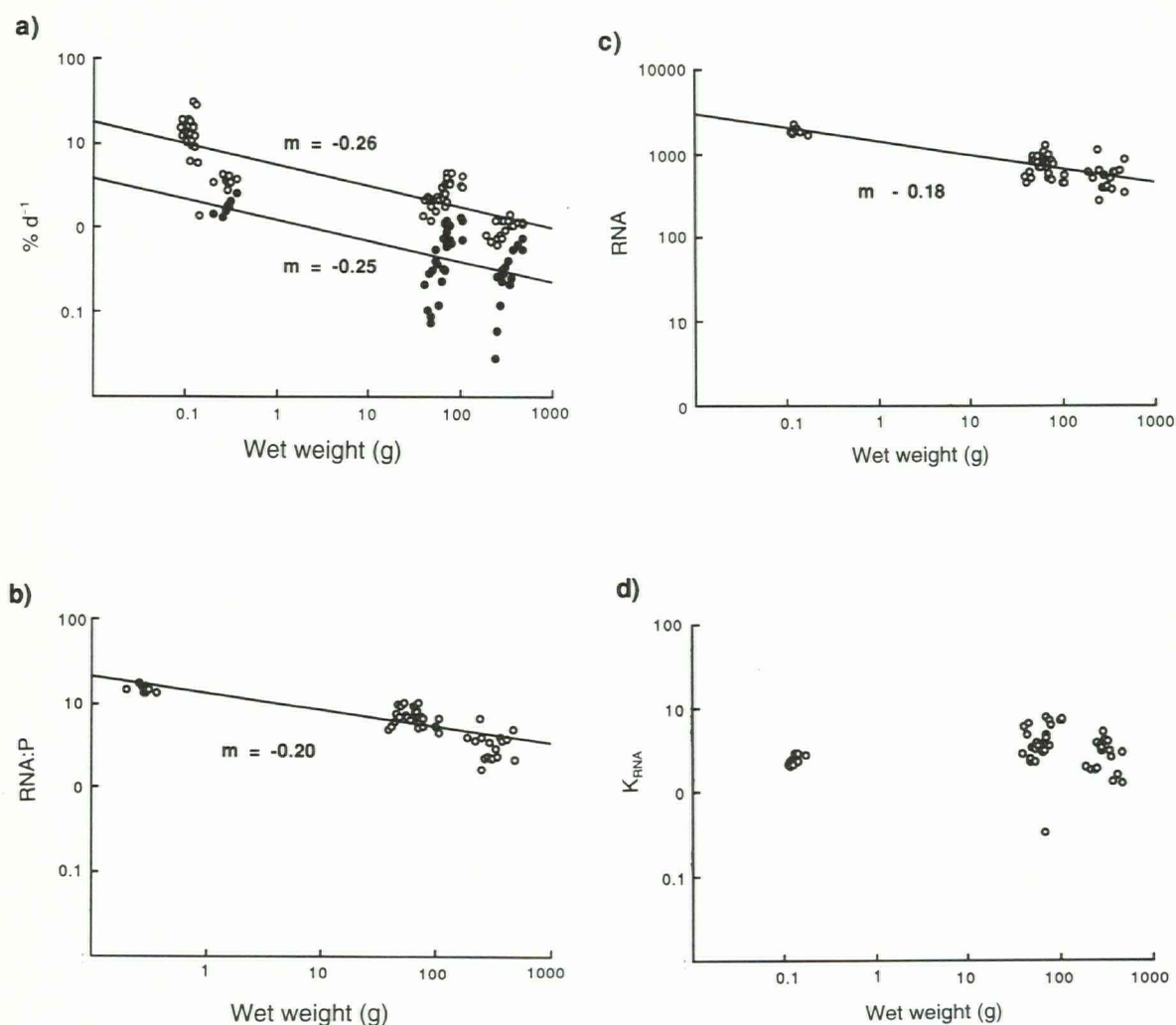


Figure 5. The relationship between body weight (g) and (a) whole-animal fractional rates of protein synthesis (k_s , % d^{-1} , open circles) and protein growth (k_g , % d^{-1} , closed circles), (b) whole animal RNA to protein ratios (RNA:P, mg RNA g protein $^{-1}$), (c) whole animal RNA concentration relative to fresh weight (RNA, mg g wet weight $^{-1}$), and (d) whole animal RNA efficiency (K_{RNA} , g protein synthesized per day per unit of cellular RNA). The data presented are for rainbow trout at 10°C, the slope for the regression line (where significant) is indicated. The regression analyses are shown in Table 3.

Table 3. Scaling equations for fractional rates of protein synthesis (k_s , % d^{-1}), protein growth (k_g , % d^{-1}), RNA to protein ratios (RNA:P, μg RNA mg protein $^{-1}$), RNA concentrations relative to fresh weight (mg RNA g wet weight $^{-1}$) and RNA efficiencies (K_{RNA} , g protein synthesized per day per unit of cellular RNA). The equations presented are for rainbow trout at 10°C and are based on the data in Figure 5.

Equation	a ($\pm$ s.e.)	b ($\pm$ s.e.)	R ²	n	p
$\log_{10} k_s \text{ v } \log_{10} \text{weight}$	0.765 (0.028)	-0.259 (0.019)	0.753	79	0.001
$\log_{10} k_g \text{ v } \log_{10} \text{weight}$	0.081 (0.089)	-0.248 (0.047)	0.364	54	0.001
$\log_{10} \text{RNA:P v } \log_{10} \text{weight}$	1.117 (0.037)	-0.204 (0.019)	0.679	54	0.001
$\log_{10} \text{RNA v } \log_{10} \text{weight}$	3.177 (0.030)	-0.184 (0.016)	0.727	54	0.001
$\log_{10} K_{RNA} \text{ v } \log_{10} \text{weight}$	0.448 (0.057)	-0.033 (0.030)	0.022	54	0.31

Table 4. Whole-animal RNA to protein (RNA:P, mg RNA g protein⁻¹) and RNA:DNA (mg RNA mg DNA⁻¹) ratios in different sizes of fish (weight, g) at various water temperatures (T°C). The data presented here are used to construct the nucleic acid ratio scaling relationships drawn in Figure 6 and the regression analysis presented in Table 5.

Species	Weight	RNA:P	RNA:DNA	T°C	Reference
Herring larvae	0.0001	46.9	4.7	5	Raae <i>et al.</i> (1988)
	0.0001	54.0	5.4	8	
Rainbow trout fry	0.148	41.0	5.2	11	Mathers <i>et al.</i> (1993)
	0.203	44.6	2.8	5	
	0.210	48.0	3.8	11	
	0.292	27.7	2.5	10	
	0.314	27.3	2.3	15	
	0.320	45.0	4.0	11	
	0.408	54.0	5.0	11	
Cod	85	5.2	1.0	14	Von der Decken and Lied (1992)
	124	6.0	1.2		
	169	5.8	1.3		
Saithe	996	8.6	1.3	7	Mathers <i>et al.</i> (1992a)
Saithe	225	6.0	0.9	12	Mathers <i>et al.</i> (1992b)
	1498	5.6	1.0		

Table 5. Scaling equations for the relationships between body weight (g) and (i) whole-body RNA:protein (RNA:P) and RNA:DNA ratios and (ii) whole-body total free amino acid (FAA) and free indispensable amino acid content (IAA) in fish. The equations are based on the data presented in Tables 4 and 6 and shown in Figures 6 and 7.

Equation	a (±s.e)	b (±s.e)	R ²	n	p
log ₁₀ RNA:P v log ₁₀ weight	1.315 (0.060)	-0.163 (0.027)	0.718	15	0.001
log ₁₀ RNA:DNA v log ₁₀ weight	0.761 (0.045)	-0.110 (0.020)	0.663	15	0.001
log ₁₀ FAA v log ₁₀ weight	1.702 (0.030)	-0.131 (0.014)	0.844	16	0.001
log ₁₀ IAA v log ₁₀ weight	1.273 (0.048)	-0.182 (0.022)	0.817	16	0.001

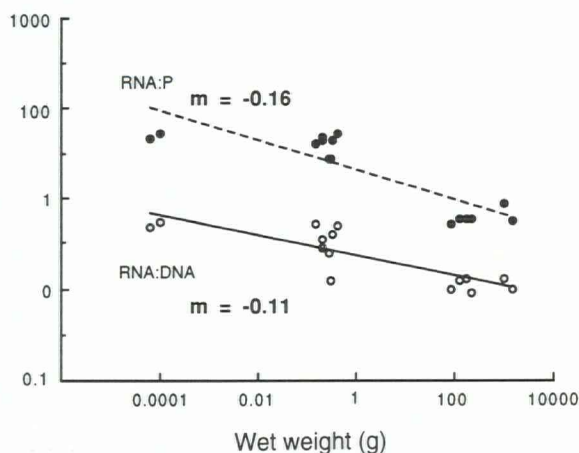


Figure 6. A summary of the relationship between body weight (g) and whole animal RNA to protein (RNA:P, mg RNA g protein⁻¹) and RNA:DNA (mg RNA mg DNA⁻¹) ratios in fish. The figure is calculated from the data presented in Table 4. The regression analyses are shown in Table 5.

phenylalanine in the fry at the end of the experiment compared with control fish. Zeitoun *et al.* (1977) reported that the free phenylalanine content of rainbow trout yolk-sac fry (minus yolk sac) is 29 nmole fish⁻¹ and the combined free phenylalanine content of the fry plus the yolk sac is 186 nmole fish⁻¹. The yolk-sac fry used in this experiment had undergone partial yolk-sac absorption (approximately 50%) and the yolk sac was not removed prior to analysis. Therefore, assuming 107 nmole fish⁻¹ is a reasonable estimate of the free phenylalanine concentration before the fry were placed in the bathing solution, the degree of flooding can be calculated as twofold above basal levels, which is not a very large expansion of the free phenylalanine pool compared with the flooding reported in other experiments (e.g., Houlihan *et al.*, 1988; McCarthy, 1993).

In Figure 2b, results obtained from bathing herring larvae (*Clupea harengus* L.) in radiolabelled phenylalanine are presented (Houlihan *et al.* 1995). The larvae (individual dry weight approximately 0.08 mg) were maintained in aerated sea water at 8°C and bathed in a 2% phenylalanine solution as in the previous experi-

Table 6. The total free amino acid (FAA, $\mu\text{mol g wet weight}^{-1}$) and free indispensable amino acid content (IAA, $\mu\text{mol g wet weight}^{-1}$) for fish of different body weights (g) at various water temperatures ($^{\circ}\text{C}$). The data presented here are used to construct the free amino acid pool scaling relationships drawn in Figure 7. The amino acid concentrations presented here are for whole-animal or white muscle (indicated by ¹).

Species	Weight	TFAA	IAA	T $^{\circ}\text{C}$	Reference
Cod eggs	0.001	200	98	5	Fyhn and Serigstad (1987)
Turbot eggs	0.001	115	65	7	Rønnestad <i>et al.</i> (1992)
Halibut eggs	0.01	100	58	7	Rønnestad <i>et al.</i> (1993)
	0.01	90	55		
Carp larvae	0.05	57	18	26	Fauconneau (1984)
Atlantic salmon yolk-sac fry	0.075	60	22	10	McCarthy <i>et al.</i> (unpubl.)
Rainbow trout fry	0.139	80	49	13	Zeitoun <i>et al.</i> (1977)
Rainbow trout fry ¹	8	32	8	15	Yokoyama and Nazakoe (1991)
Sea bass ¹	8	41	11	18	McCarthy (1993)
Rainbow trout ¹	70	29	7	12	Kaushik and Luquet (1979)
Rainbow trout ¹	70	44	10	12	Kaushik and Luquet (1979)
Rainbow trout ¹	175	22	4	12	Gras <i>et al.</i> (1978)
Rainbow trout ¹	265	32	17	11	Carter <i>et al.</i> (1995)
Cod ¹	322	16	9	11	Lyndon <i>et al.</i> (1993)
Atlantic salmon ¹	425	32	6	7	Carter <i>et al.</i> (unpubl.)
Catfish ¹	933	15	5	?	Wilson and Poe (1974)

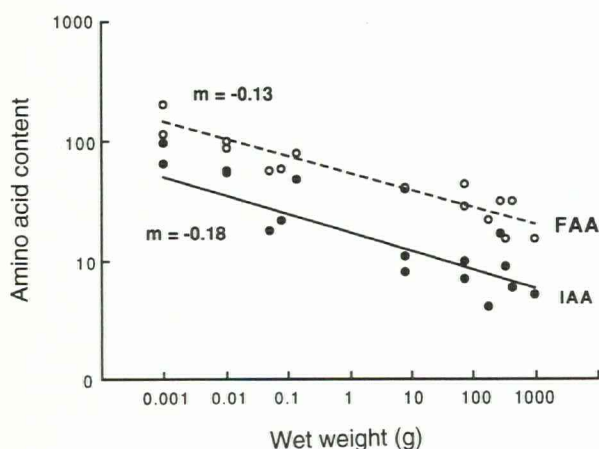


Figure 7. A summary of the relationship between body weight (g) and the total free amino acid (FAA, $\mu\text{mol g wet weight}^{-1}$) and free indispensable amino acid content (IAA, $\mu\text{mol g wet weight}^{-1}$) in fish. The figure is calculated from the data presented in Table 6. The regression analyses are shown in Table 5.

ment. At various times after the addition of the phenylalanine, groups of fish were removed, treated in a similar fashion as the yolk-sac fry, and the free phenylalanine and protein-bound specific radioactivities determined as described previously. The free phenylalanine-specific radioactivity reached high stable values within 3 h of the beginning of the incubation and remained elevated for up to 13 h. The average free phenylalanine-specific radioactivity over the time course, $920 \text{ dpm nmole}^{-1}$, was approximately 75% of the specific radioactivity of the bathing medium.

These data on rates of uptake of radiolabelled amino

acids by herring larvae in sea water are similar to those obtained with juvenile nase (Houlihan *et al.*, 1992) and tilapia (Houlihan *et al.*, 1993b) and are in contrast to those described above for rainbow trout yolk-sac fry. Dissolved organic substrates are absorbed by fish across the gills, digestive tract, and skin (Fauconneau *et al.*, 1989). Low molecular weight compounds may be selectively reabsorbed by gill chloride cells (Furspan and Isaia, 1983) and it has been suggested that this capacity for substrate reabsorption could facilitate the intake of radiolabelled amino acids into the fish (Fauconneau *et al.*, 1989). Both the herring larvae described in the above experiment and the studies on nase and tilapia (Houlihan *et al.*, 1992, 1993b) were carried out using active feeding animals and it is likely that phenylalanine was absorbed from the water by the three routes described above. In the marine larvae the animals may be mainly taking up the phenylalanine when they are drinking. The low level of flooding of the free amino acid pool in the rainbow trout may be due to its early developmental stage. In the yolk-sac fry the mouth is still closed, the gut is not fully developed and therefore the surface area for uptake may be reduced to the skin and the gills.

Injecting juvenile fish

Small fish present problems for injection of radiolabelled amino acids or other substances because of the small amounts to be injected if the ratio of volume injected to live weight (1:100) used in larger animals (e.g., Houlihan *et al.*, 1988) is to be maintained. In Figure 2c rainbow trout yolk-sac fry (110 mg wet weight, 10°C , McCarthy *et al.*, unpubl.) were injected with $1.5 \mu\text{l}$

of 135 mM phenylalanine/L-[2, 6- ^3H] phenylalanine per fish into the yolk sac using a micro-injector (Zhu *et al.*, 1985). The phenylalanine-specific radioactivity of the injection solution was 930 dpm nmole $^{-1}$. Larger feeding rainbow trout fry (203 mg wet weight, 5°C, Mathers *et al.*, 1993) were injected into the peritoneum with a similar injection solution using a Hamilton syringe. The volume injected was 1 μl 100 mg $^{-1}$ wet weight and the phenylalanine-specific radioactivity of the injection solution was 1600 dpm nmole $^{-1}$. This approach permitted injection into the yolk sac or the peritoneum without damaging the fry. In both experiments the fish were placed in small, well-aerated tanks at the appropriate water temperature and fish were removed at 0.5, 1, and 2 h after injection. Treatment of the samples to determine the whole-body free pool and protein-bound phenylalanine-specific radioactivity are described in Houlihan *et al.* (1986, 1988) and the mean values for each time interval are shown in Table 1. In both groups of fish the free phenylalanine-specific radioactivity remained elevated and stable over the time course (Fig. 2c). However, the free phenylalanine-specific radioactivity of the feeding fry was higher, attaining a mean value of 80% of the injection solution compared to the yolk-sac fry, where the free-pool-specific radioactivity was 30% of the injection solution. The dynamics of yolk-sac absorption remain unclear and it is not known whether free amino acids are absorbed uniformly from the yolk sac during development or whether there are specific sites for absorption. If absorption of free amino acids is limited to specific site then this may also help to explain the low flooding rates in the yolk-sac fry.

Rates of protein labelling and protein synthesis achieved using the bathing and injection methods

The rate of labelling of proteins in the yolk-sac fry of rainbow trout bathed in radiolabel is shown in Figure 3a and appears to plateau after 6 h incubation. The fractional rate of protein synthesis (k_s) was calculated using the equations of Fauconneau (1984), assuming that the specific radioactivity of the protein-bound phenylalanine reached a plateau value with a single rate constant. The calculated rates of protein synthesis are shown in Table 1 and the average synthesis rate over the incubation time in the yolk-sac fry was 164% d $^{-1}$. In Figure 3b the rate of labelling of the phenylalanine in the proteins of larval herring from the experiment described in Figure 2c is presented. The protein-bound phenylalanine-specific radioactivity increased linearly as in many previous reports (e.g., Houlihan *et al.*, 1992; Carter *et al.*, 1993) and from linear regression analysis it can be estimated that the proteins began to be labelled 67 min

after the introduction of the radiolabelled phenylalanine into the water. Calculation of the rate of protein synthesis can therefore be performed after subtracting 67 min from the protein-bound phenylalanine-specific radioactivities at different times during the course of the experiment (Houlihan *et al.*, 1995). As the free phenylalanine-specific radioactivity remained constant over the incubation time, fractional rates of protein synthesis were calculated using the equation of Garlick *et al.* (1983). The significance of the protein synthesis rates relative to protein growth rates is discussed below.

The rates of protein labelling in the yolk-sac fry and small feeding fry injected with ^3H -phenylalanine (Fig. 3c) were linear over the incorporation time with intercepts that were not significantly different from zero ($p < 0.46$ and $p < 0.34$ respectively). In Figure 2c, the free phenylalanine-specific radioactivity for the two groups of fish either remained stable (yolk-sac fry) or showed a slow linear decline (small fry) over the incorporation time. Therefore fractional rates of protein synthesis could be calculated using the equation of Garlick *et al.* (1980), which assumes a slow linear decline in the free-pool-specific radioactivity with time, or Garlick *et al.* (1983), which assumes a stable free pool over the incorporation time.

Fractional rates of protein synthesis in whole rainbow trout yolk-sac fry measured using the flooding dose injection (18% d $^{-1}$) were considerably lower than the values obtained using the bathing method of Fauconneau (1984). Early studies on fish larval protein synthesis have suggested that protein degradation accounts for between 90 and 95% of the protein synthesized per day (Fauconneau, 1984; Fauconneau *et al.*, 1986a, b). Although the energetic costs of protein degradation are not fully understood, several energy consuming pathways have been identified (Hawkins, 1991). Recent studies have shown that these high rates of protein turnover would be energetically impossible in larval fish (Wieser *et al.*, 1988; Wieser and Medygsey, 1990; Wieser, 1991), and therefore the results obtained using the bathing method of Fauconneau (1984) must be treated with some caution. The ease with which the flooding dose technique by injection can be applied to measuring synthesis rates in larval and small juvenile fish will enable the measurement of protein turnover to be incorporated into future studies of growth and energetics of larval and small juvenile fish.

Amino acid flux model for larval herring

A model of the amino acid flux in a larval herring of 0.1 mg dry weight is shown in Figure 4. The protein consumption rate is taken as 11% of the body protein content (Checkley, 1984; Houlihan *et al.*, 1995) and

assimilation efficiency is assumed to be 79% (Pedersen and Hjelmeland, 1988; Kiørboe, 1989). The total free amino acid pool concentration was taken from direct measurement of first-feeding larval herring. The rate of protein synthesis of about $13\% \text{ d}^{-1}$ for a larva growing at $7\% \text{ d}^{-1}$ was calculated from the relationship between growth rate and protein synthesis (Houlihan *et al.*, 1995). Fractional rates of protein degradation, estimated from the difference between rates of protein synthesis and growth, were approximately $6\% \text{ d}^{-1}$. Nitrogenous excretion was calculated as rates of amino acid equivalents and assumed to be around 20% of the absorbed dietary protein.

The value of this model is that, although there are a number of uncertainties which further experimental work will correct, it does point to the differences between small and large fish already described in Figure 1. Instead of the daily dietary amino acid intake representing almost twice the whole animal free amino acid pool, as it does in larger fish, in small herring larvae the amino acid intake is only one-third of the whole animal's free amino acid pool (Fig. 4). This might imply that the arrival of the ingested amino acids would have a relatively small effect on the free amino acid pool composition. However, almost half of the total free amino acid pool will be used for protein synthesis. We know little about how the free amino acid pools change during the course of a day in larval fish. In larger fish we have suggested above that the free pools remain relatively constant following a meal. What may be important is that the ratio between the total free pool and the whole-animal protein-bound amino acids is around 0.3 in larval herring but 0.02 in larger fish (Fig. 1). Thus, in larval fish there is little scope for re-cycling of essential amino acids if these become limiting in the diet.

RNA concentrations

Another way of estimating rates of protein synthesis is to use the database available relating RNA concentrations and fractional rates of protein synthesis which permits calculation of RNA efficiencies (Millward *et al.*, 1973; McMillan and Houlihan, 1992). If the RNA concentration of larval fish is determined then maximum likely rates of protein synthesis can be estimated. It has recently been proposed that in mammals it is very rare for RNA translational efficiency to exceed 20 g protein synthesized per day per unit of cellular RNA (Reeds and Davis, 1992). In growing fish most values for RNA efficiencies are around 3 to 6 g protein synthesized per day per unit of cellular RNA (Table 2). In larval herring, whole-body RNA to protein ratios of $40 \mu\text{g RNA mg}^{-1}$ protein and RNA efficiencies of up to a maximum of 6 have been observed (Houlihan *et al.*, 1995). If the value

of $40 \mu\text{g RNA mg}^{-1}$ protein is accepted and a likely maximum RNA efficiency of 6, then fractional protein synthesis rates are likely to be around $24\% \text{ d}^{-1}$. This value is close to the results actually obtained by the bathing technique (see above). Maximum growth rates of larval herring for the size under consideration have been reported to be around $8\% \text{ d}^{-1}$ (Kiørboe *et al.*, 1987). Thus, the ratio of growth to synthesis from these considerations is around 33%, which agrees with the directly determined values reported for the herring model. Admittedly these calculations are open to a number of criticisms, not the least of which is the uncertainty in the appropriate RNA efficiency value. This is known to vary considerably in the short term; after a meal, the RNA efficiencies may be high and RNA concentrations are variable depending upon the feeding level (McMillan and Houlihan, 1989). However, in the above calculations the maximum RNA efficiencies have been used for a given temperature (Table 2), lower efficiencies would suggest lower protein synthesis rates and therefore higher efficiencies of retention of synthesized proteins. The question of body size effects may also be important.

Scaling relationships

The measurement of various physiological and biochemical parameters in larval fish provides the opportunity to test the hypothesis that larval fish follow the same scaling relationships that have already been established for larger fish. Thus, a number of questions, raised in the previous sections concerning rates of protein synthesis, free amino acid pool concentrations and RNA concentrations in larval fish, may be viewed from the perspective of body size. In this section these relationships are examined in two different ways, firstly using data available from our laboratory for four different sizes of rainbow trout at 10°C (McCarthy *et al.*, unpubl.) and, secondly, using data available from the literature for various species of fish. The two data sets were analysed assuming an allometric relationship between the appropriate variable (Y) and weight (X) such that $Y = a \times X^b$, where a is a constant and b the weight exponent. As is usual in this kind of analysis, the relationship between weight (X) and the variable is expressed in the form: $\log Y = \log a + b \times \log X$, and the Y value may be expressed relative to weight as in fractional rates of protein synthesis. For rainbow trout at 10°C fractional rates of protein synthesis and growth both decrease with weight with similar exponents (Fig. 5, Table 3). This has been proposed for individual tissues of rainbow trout (Houlihan *et al.*, 1986), but the new data presented here are for whole animals and are also over a larger range of body weights. Although the frac-

tional rates of protein growth and synthesis are parallel with increasing body size, the measured synthesis retention efficiencies (k_g/k_s) decline with increasing body weight ($50.50 \pm 11.52\%$ for a 0.162 g fry and $24.56 \pm 2.84\%$ in an animal of 316 g).

However, to be certain of this apparent decline in retention efficiencies the data would need to include protein growth rates over the same range as the synthesis rates. Therefore we still cannot show conclusively that larval synthesis retention efficiencies are high in small fish. In fact, the available data suggest that synthesis retention efficiencies are similar for all sizes of fish (e.g., Houlihan *et al.*, 1988, 1992, 1993b; Carter *et al.*, 1993; McCarthy, 1993; Mathers *et al.*, 1993).

Previously, the slope of the relationship between body weight and fractional rate of protein synthesis had been thought to be much steeper than that suggested here (-0.77 in Dabrowski (1986) compared with -0.259 , Table 3). The steep slope was a result of the high protein synthesis rates that were thought to be characteristic of juvenile fish. The slope values suggested in the present study are much more conventional and similar to the decline in the weight-specific rates of oxygen consumption (-0.2 , Rao, 1968; Reiss, 1989), growth (-0.35 to -0.40 , Jobling, 1992), and ammonia excretion (-0.434 , Dabrowski, 1986) reported for fish.

The whole-animal RNA to protein ratio ($\mu\text{g RNA mg protein}^{-1}$) gives an approximate indication of ribosomal number (Millward *et al.*, 1973; Pain and Clemens, 1980) and has been termed the capacity for protein synthesis (Preedy *et al.*, 1988). From the rainbow trout scaling data and the data available in the literature it appears that the capacity for synthesis decreases with size due to a decrease in the concentration of ribosomes (Figs. 5b, c, 6, Tables 3, 4, and 5). An alternative measure of RNA concentration is the RNA to DNA ratio (RNA:DNA, $\mu\text{g RNA } \mu\text{g DNA}^{-1}$) and this has a lower slope than the RNA to protein ratio and RNA to fresh weight ratio ($\text{mg RNA g wet weight}^{-1}$).

We suggest that the concentration of RNA is the primary factor controlling rates of protein synthesis in the tissues and this is best expressed as RNA to protein ratios or RNA to fresh weight or dry weight (Mathers *et al.*, 1992a, b). As RNA concentrations in the tissues decline with increasing body size, fractional rates of protein synthesis also decline, resulting apparently in the RNA efficiency being independent of weight (Fig. 5d). These results emphasize the previous discussion where high rates of protein synthesis in larval fish were ascribed to the high concentrations of RNA.

It is also very interesting that the concentrations of total free amino acids and indispensable free amino acids in the tissues of fish decline with increasing body size (Table 6, Fig. 7) with exponents of -0.13 and -0.18 respectively (Table 5). As these exponents are similar to

those for fractional rates of protein synthesis and the RNA to protein ratio, it is tempting to suggest that the two are related i.e., high rates of protein synthesis require a large free pool. A negative correlation between body size and total amino acid free pool concentration has been found in mammals, but it has been suggested that similar relationships do not hold for individual amino acids (Waterlow *et al.*, 1978).

For a given size of fish, each of the variables discussed above will be influenced by a variety of environmental factors, such as temperature or food intake. To minimize the effect of nutritional plane the data presented are for growing fish. The relationships appear robust enough to absorb variation due to species and temperature differences, although a recommendation of our analysis would be to construct allometric relationships for each species using fish held at the same temperature.

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